

Checkpoint 8

Data 6560 Sport Analytics
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Risk vs. Reward: Analyzing Aggressive vs. Conservative Decision-Making in Golf

Primary Stakeholder: Competitive golfers, coaches, and performance analysts

Problem and Decision

Golfers constantly face a strategic choice: play aggressively to gain strokes or play conservatively to avoid mistakes. This project evaluates whether a more aggressive ball striking style actually leads to better scoring outcomes at the professional level.

The decision this analysis informs is:

Should players and coaches encourage a more aggressive ball striking strategy, or does conservative play produce more reliable results?

A good decision here improves tournament scoring, increases cut making probability, and raises finishing position consistency.

Data in One Line

Dataset: Cleaned PGA Tour strokes gained data (Checkpoint 4)

ASA_All_PGA-Tourn_Level_Derived.csv

Row definition: One player per tournament

Time span: 2022 PGA Tour season, with focused analysis on the 2022 Masters

Scope: 29,180 player-tournament observations (full season) and 78 players (Masters only)

Approach in One Line

I used linear regression and group comparisons to estimate how an aggressiveness proxy – ball striking aggressiveness (SG Off the Tee + SG Approach) – relates to total strokes gained, controlling for putting performance.

This approach isolates how aggressive ball striking translates into actual scoring outcomes.

Key Result

Players in the most aggressive ball striking group gain about 3.5–4.0 more strokes per tournament than the most conservative group.

- Full 2022 season:
 - High-aggressiveness players averaged **+1.26 SG_Total**
 - Low-aggressiveness players averaged **-2.22 SG_Total**
- Masters 2022 results showed the same pattern, even in an elite, high pressure field.

This effect is large enough to meaningfully change tournament finishes, cut outcomes, and earnings.

One Key Exhibit

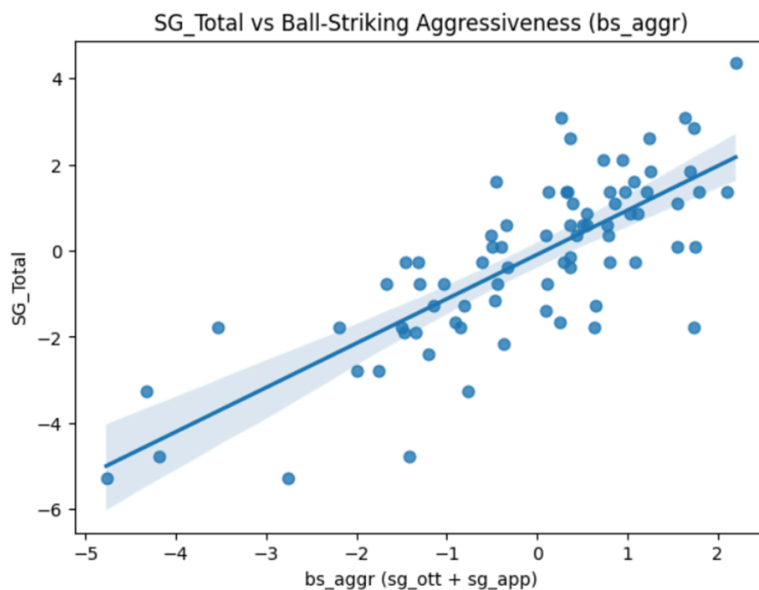


Figure 1: Total Strokes Gained vs Ball Striking Aggressiveness (2022 Masters)

Takeaway: Highly aggressive ball strikers consistently outperform conservative players by several strokes gained per tournament.

Recommendation

Golfers and coaches should encourage controlled aggressiveness in ball striking, particularly off the tee and on approach shots.

How to act on this:

- Prioritize practice and strategy that improves aggressive tee shots and aggressive approach shots, even if it increases variance.

- Accept some downside risk in exchange for higher upside scoring.
- Track success using total strokes gained and finishing position, not just fairways hit or greens in regulation.

When: Immediately, as this approach applies across both regular PGA Tour events and majors.

Risks and Limits

1. Aggressiveness is measured indirectly. The analysis uses strokes gained components, not shot by shot decision data. Some aggressive outcomes may reflect skill rather than intentional risk taking.
2. Course context matters. Augusta National and similar courses may reward aggressiveness more than others. Strategy should still be adapted to course setup and conditions.

Mitigation:

Monitor performance by course type and re-evaluate aggressiveness thresholds on narrow or penalizing layouts.

Next Steps

1. Refine the aggressiveness metric by replacing the combined measure with separate SG_OTT and SG_APP thresholds to tailor strategy more precisely.
2. Extend to prediction by using the model to predict cut making probability or Top 20 finishes based on aggressiveness levels.
3. Decision level analysis by incorporating shot by shot data to directly compare laying up vs. going for green decisions.